
RangeViT: Towards Vision Transformers for 3D Semantic Segmentation in Autonomous Driving

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(CVPR 2023)

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LiDAR: Point Clouds

Raw Data: A "Point Cloud" of (x, y, z, i) coordinates.

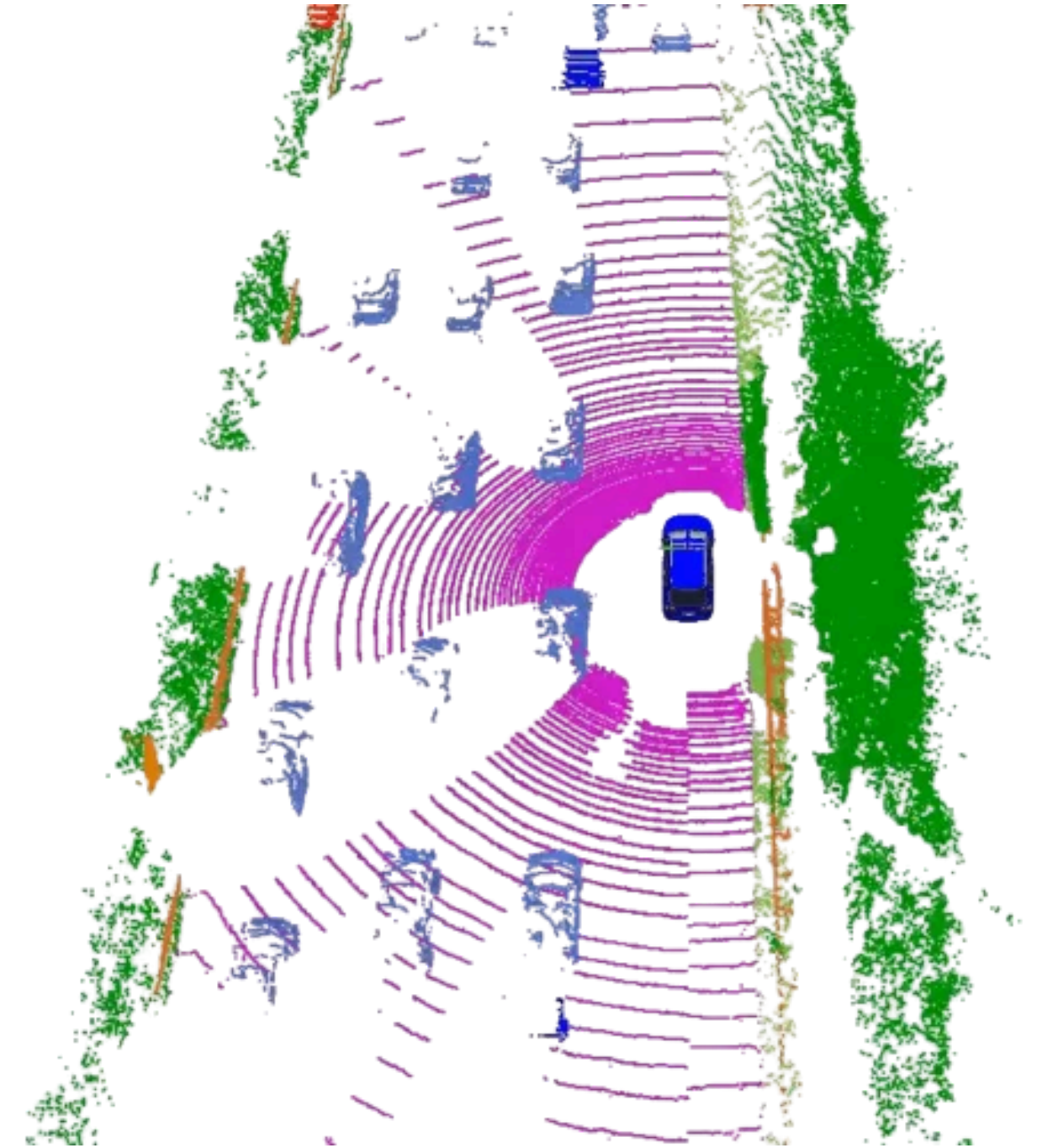
- i : intensity
- (x, y, z) : Cartesian coordinates

Datasets: SemanticKitti (64 laser beams, 19 classes)
21 sequences each with 4400+ point cloud images

Training: 00-07, 09, 10

Validation: 08

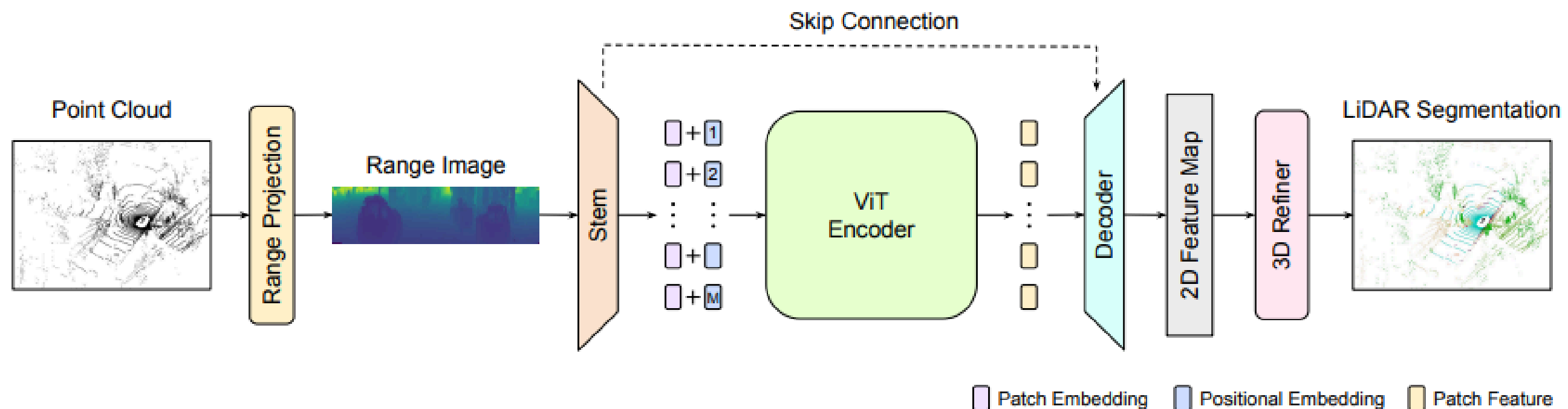
Testing: 11-21



RangeViT Architecture

Baseline mode:

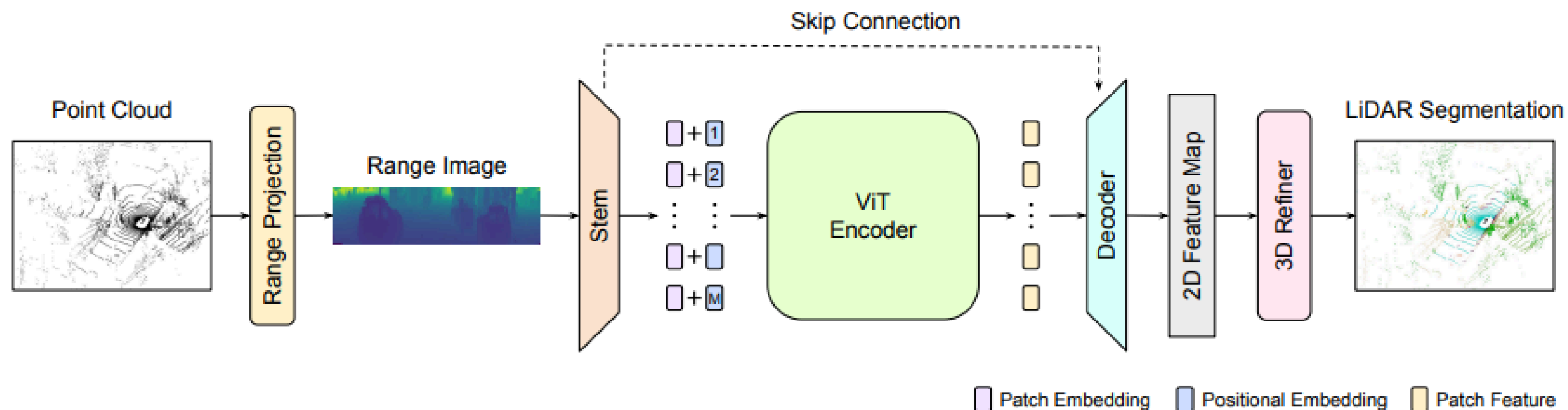
- **Input:** (x, y, z, i) raw 3D point cloud.
- **Conv:** captures local spatial relationships aka inductive bias.
- **ViT Encoder:** Long-term dependencies
- **Decoder:** Uses Pixel Shuffle technique that rearranges deep data into sharp pixels.



RangeViT Architecture

Baseline mode:

- **3D Refiner:** converts the 2D image to 3D and fixes overlaps that might occur during previous steps, and bilinear upsampling.
- **Final logits:** Probabilities of each pixel for each class.
- **The Result:** A 5-channel image (r, x, y, z, i). It looks like a long, panoramic photo where “ r ” represents the distance from the sensor.



Novelty 1: Interpretable constraints

A. Prototype-Only Objective

- Added class prototypes as a regularizer in training.
- Prototypes influence loss only, not the forward prediction path.
- **Purpose:** Test whether cluster-structure alone helps.

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B. Geometry-aware prototype

- Extended prototype space from feature-only to **feature + geometry**.
- Decoder features are concatenated with LiDAR geometry channels for prototype computations.
- **Pull features** of the correct class toward their prototype. This makes same-class features cluster tightly.
- **Separation:** Push features away from wrong-class prototypes by a margin, reducing confusion between similar classes.

Novelty 1: Interpretable constraints

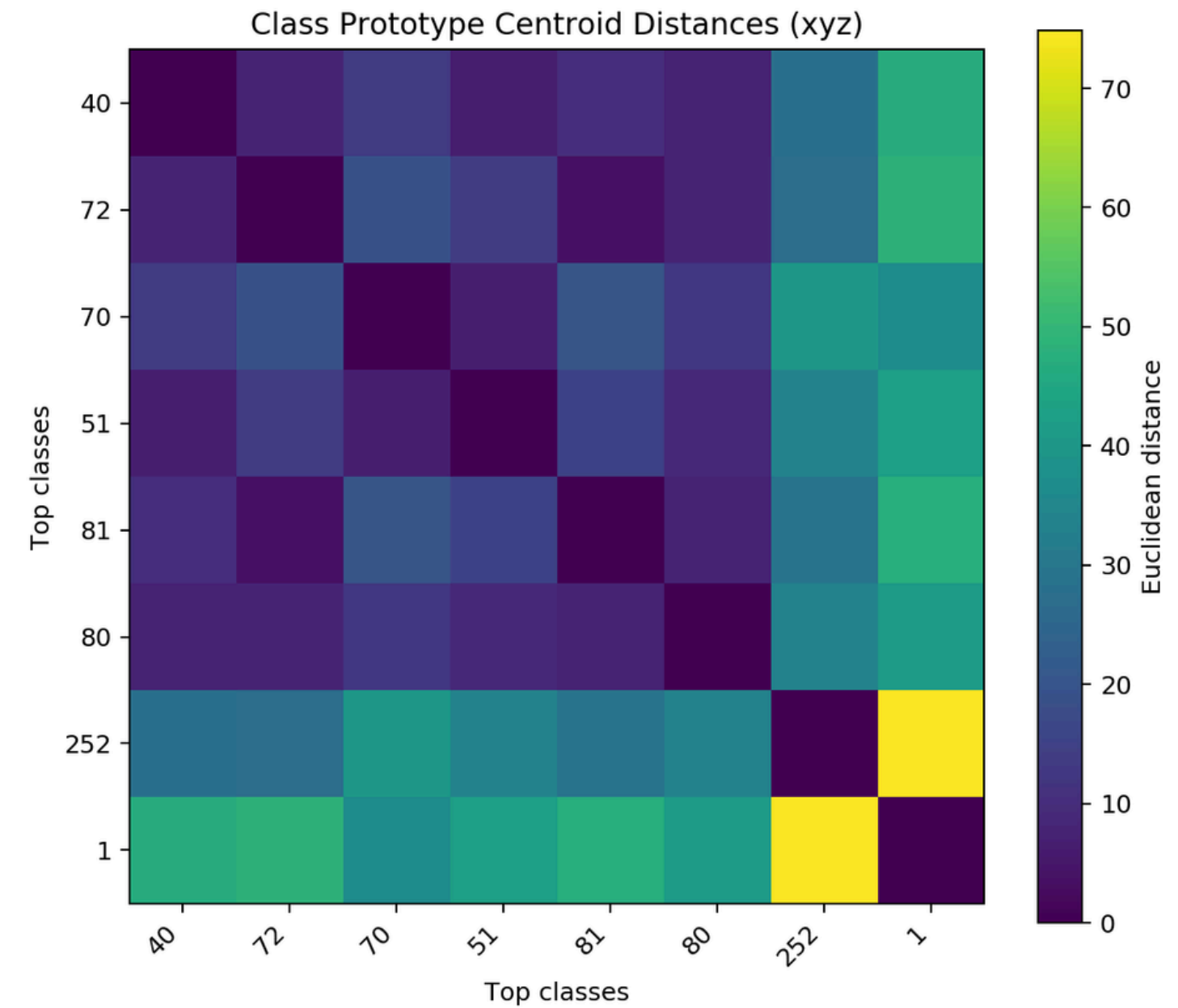
C. Attention Entropy Regularization

- Exposed transformer attention maps during forward.
- Added entropy regularisation toward a target entropy.
- **Purpose:** Avoid attention collapse or over-uniform attention and stabilise saliency behavior.

Ablation analysis: $\text{New Loss} = L_{\text{seg}} + \lambda_{\text{proto}} * L_{\text{proto}} + \lambda_{\text{geo}} * L_{\text{geo}} + \lambda_{\text{attn}} * L_{\text{attn}}$

Method	Acc	mIoU	Recall	Δ mIoU vs Baseline
Baseline stable	0.4668	0.3629	0.4746	0.0000
Proto-only stable	0.4481	0.2767	0.3506	-0.0861
Geo-proto	0.4348	0.3861	0.4394	+0.0233
Full objective (geo-proto + attention reg, no fusion)	0.4425	0.3922	0.4413	+0.0294

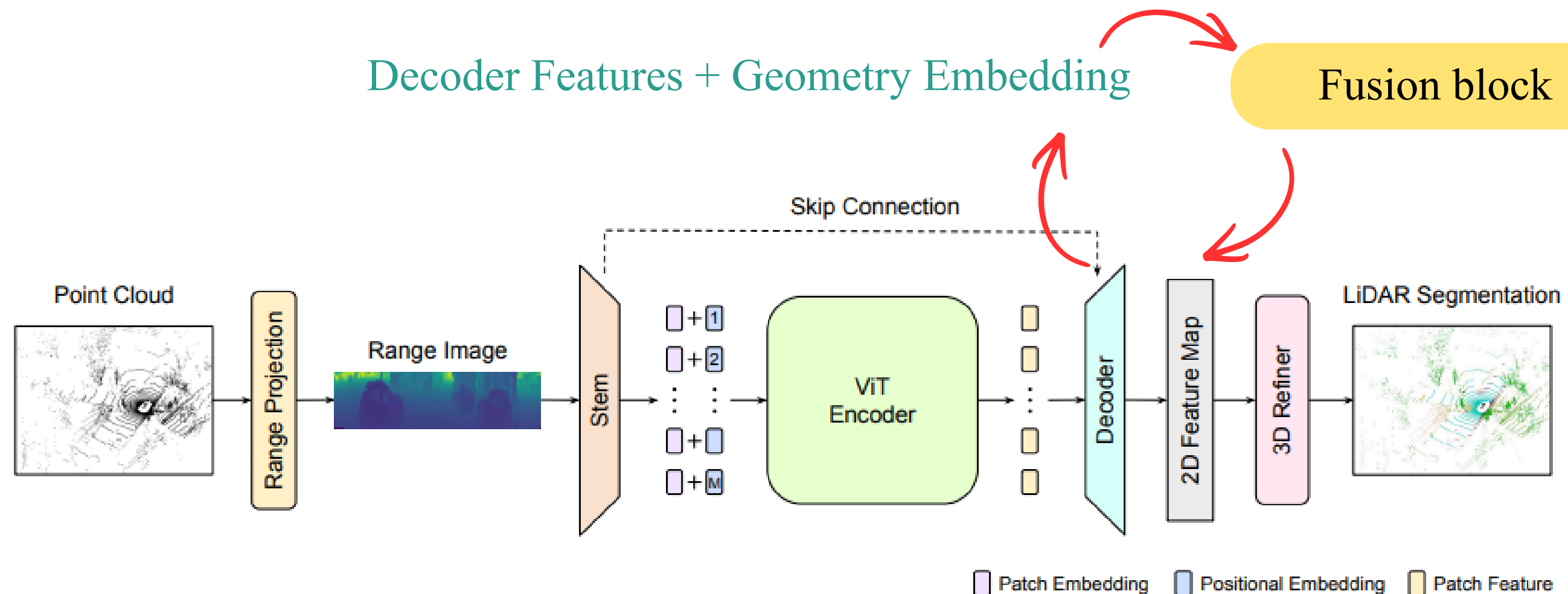
Centroid distance heatmap



Novelty 2: Geo Fusion Module

Geometry Fusion Module

- Input geometric channels are projected with a 1x1 conv into a compact geometry embedding.
- Decoder features and geometry embedding are concatenated; 3x3 fusion conv produces fused features.
- **Purpose:** Makes the decoder more geometry-aware and improves segmentation quality.



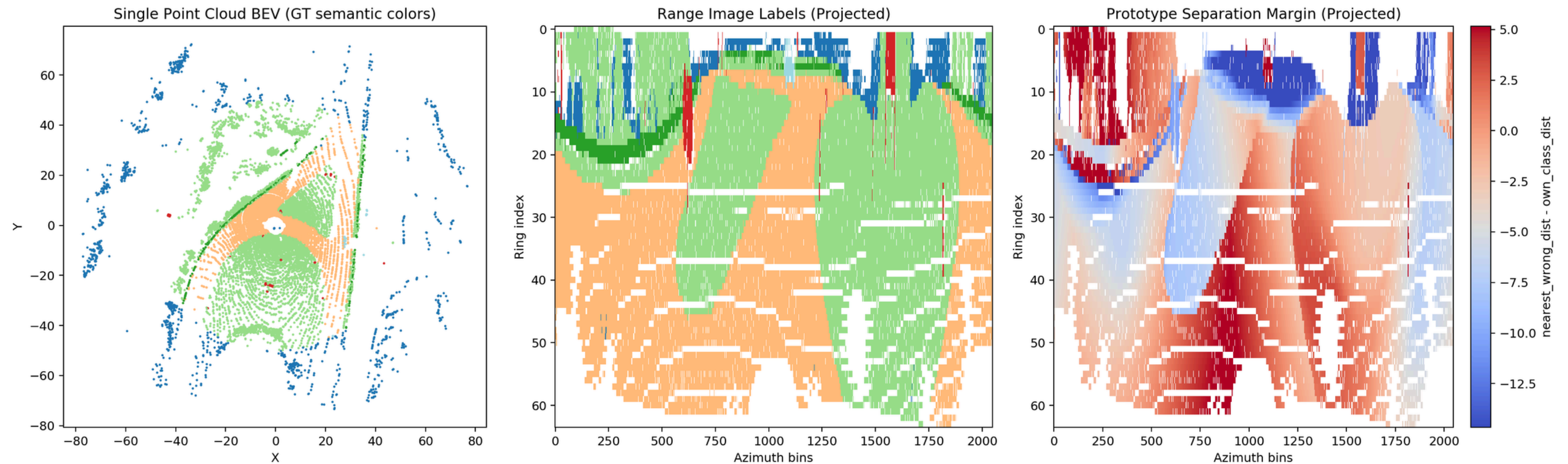
Final Results

Ablation analysis of all variants

Variant	Acc	mIoU	Recall	Δ mIoU vs Baseline
Baseline stable	0.4668	0.3629	0.4746	0.0000
Proto-only stable	0.4481	0.2767	0.3506	-0.0861
Geo-proto	0.4348	0.3861	0.4394	+0.0233
Full objective (no fusion)	0.4425	0.3922	0.4413	+0.0294
Geometry fusion only	0.5634	0.4797	0.5265	+0.1168
Geometry fusion + full objective	0.4525	0.3999	0.4476	+0.0371

Final Results Cont.

Interpretability on One Scan | Seq 01 Frame 000000.bin | Higher margin means clearer prototype-based class separation



Conclusion & Future Work:

Conclusion

- Geo Fusion improves the representation by adding explicit geometric context late in the network.
- Prototype-based interpretability makes the model explainable in class-space.
- Ablation shows prototype-only is insufficient, but prototype + geometry is effective; adding full regularization achieves the best IoU.

Future Work

- **Improve prototype learning stability:** Use adaptive prototype updates and class-balanced sampling so rare classes do not get noisy or collapsed prototypes.
- **Better geometry-aware fusion:** Move from single-stage fusion to multi-scale fusion (early + mid + late) so geometry helps both representation learning and final boundary refinement.
- **Temporal consistency across scans:** Extend prototype and margin reasoning from single scan to sequence-level consistency for moving objects and dynamic scenes.

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Thank You!